Checklist Set 1

Gamified Virtual Reality Learning Platform for the Sri Lankan School Curriculum with History

(Mathematics Component)

Project ID: 25-26J-423

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Executive Summary

The Mathematics component of **Lakmaga** – a gamified virtual reality learning platform for Sri Lanka's G.C.E. Ordinary Level curriculum – is evaluated in this feasibility study. This component addresses a critical educational problem: traditional mathematics teaching in Sri Lanka often relies on rote learning and abstract lectures with minimal interactive practice, leading to low student engagement and poor understanding of key concepts. The proposed solution is an immersive **VR Mathematics module** that allows students to learn through **interactive historical scenarios** aligned with their syllabus, making math education more engaging and practical. Key features envisioned include a *Sluice Gate Puzzle* for fractions, a *Granary Area Challenge* for geometry and division, and a *Stability and Balance Challenge* for logical reasoning with measurements – all designed to enhance understanding by connecting math problems to ancient Sri Lankan agricultural technology and culture through immersion and gamification.

This study examines the project's technical, operational, market, and financial feasibility. The findings indicate that the project is technically viable: using the Unity game engine and standalone VR headsets (e.g. Meta Quest) to create interactive 3D environments is feasible and has been demonstrated by prototypes of core features in similar educational VR projects. Operationally, the module can be integrated into schools with proper teacher training and resource planning, leveraging bilingual (Sinhala/English) content to fit the local context and curriculum. The market feasibility is strong given the lack of localized educational VR content for math in Sri Lanka and the large number of O/L mathematics candidates annually (over 340,000 in 2022), though adoption will depend on support for hardware infrastructure and teacher readiness. The financial analysis suggests moderate initial costs (for development and VR equipment) that are justified by the potential educational benefits – such as improved student performance and engagement. VR learning tools have been shown to increase STEM course pass rates by building student skills and confidence, indicating a positive return on investment in terms of educational outcomes.

Overall, the Mathematics VR component is **feasible** and has the potential to significantly improve math learning in Sri Lankan schools. It is recommended to proceed with development and pilot deployment, while addressing key risks (such as ensuring device availability, teacher training, and alignment with exam requirements). With these measures and careful planning, Lakmaga's mathematics component can become a sustainable and impactful educational innovation that transforms math education from a static, abstract experience into an interactive journey rooted in Sri Lankan heritage.

Contents

Introduction

Sri Lanka's secondary education system faces persistent challenges in teaching mathematics effectively. Lakmaga is a collaborative project aimed at enhancing learning through a gamified virtual reality platform, bringing subjects to life with historical context. This feasibility analysis focuses on the mathematics component of Lakmaga, which targets Grade 10–11 students (O/L mathematics syllabus). The initiative was conceived to address gaps in traditional math instruction particularly the lack of interactivity and real-world application in math classes. Currently, many schools rely on textbook-centric, teacher-driven lessons that make mathematics highly abstract and disengaging for students. There is little opportunity for students to explore mathematical ideas through tangible experiences, leading to poor conceptual understanding and high failure rates. In recent years, over one-quarter of O/L candidates have failed Mathematics, reflecting a serious deficiency in learning outcomes (in some regions and years, failure rates are reported even higher). This not only impedes students' academic progress but also signals a need for innovative teaching approaches.

Background: Integrating Virtual Reality into education is a rising global trend, proven to boost student engagement and understanding by enabling immersive, interactive learning experiences. VR creates a simulated environment where learners can visualize and manipulate objects, which is particularly beneficial for abstract subjects like mathematics. It supports spatial reasoning, long-term memory retention, and motivation by allowing students to **"learn by doing"** rather than by rote. In the context of Sri Lanka, where conventional math teaching often fails to contextualize concepts, VR offers a unique opportunity: students can experience mathematical principles through scenarios drawn from the country's own history and everyday life. This approach aligns with culturally responsive pedagogy, using familiar contexts to make learning more relatable. Moreover, many schools in Sri Lanka have limited facilities for math lab activities or practical demonstrations. A VR-based solution can provide virtual hands-on experiences (for example, exploring geometric shapes in a virtual workshop or simulating measurements in a historical setting) that are otherwise absent from the curriculum. By aligning these experiences with the national curriculum and presenting them bilingually (Sinhala and English), the platform ensures accessibility and relevance to local students.

Objective: The objective of this feasibility study is to critically evaluate the viability of implementing the Lakmaga Mathematics VR module as an individual research project and eventual educational tool. This involves assessing whether the proposed solution can be successfully developed and integrated from multiple perspectives: technical implementation, operational deployment in schools, market demand/impact, and financial sustainability. The study aims to determine if the Math component is **feasible as a standalone module** that achieves its educational goals, and to identify any conditions or recommendations for its successful execution. Key questions include: *Can the required VR technology and content be developed with available tools and expertise? Will schools, teachers, and students be able to adopt and use this solution* *effectively? Is there sufficient demand and support for a VR-based math learning tool in the target educational context? What are the cost implications versus the expected benefits?* By answering these questions, the study will inform whether and how the project should proceed.

Scope: This report covers the feasibility aspects of the **mathematics component** of the Lakmaga VR learning platform. It addresses core mathematical topics in the O/L syllabus – such as fractions, algebra, geometry, measurements, and logical problem-solving – delivered via VR experiences set in ancient Sri Lankan contexts. The scope includes an analysis of the current problems in math education, a description of the proposed VR solution, and detailed feasibility evaluations (technical, operational, market, financial) for implementing the solution. Wider project aspects like the History or Science components of Lakmaga are beyond the scope, except for how they inform the interdisciplinary and cultural approach. The focus remains on Grade 10–11 mathematics learning and the local schooling environment. Both the capabilities of modern VR technology and the constraints of Sri Lankan school infrastructure are considered in defining this scope.

Problem Statement

Mathematics education in Sri Lanka's secondary schools is facing a **serious engagement and achievement crisis**. Several underlying issues have been identified:

- **Lack of Engagement and Practical Context:** Traditional teaching methods for math are largely "chalk-and-talk," focusing on lectures, note-taking, and rote problem drills. Abstract concepts like algebraic fractions or geometric theorems are taught on the blackboard with minimal real-life context or visualization. Students often struggle to grasp these ideas because they cannot relate formulas and diagrams to tangible experiences. There is a noticeable absence of interactive learning aids or lab activities in math compared to other subjects. As a result, many students find math uninteresting and irrelevant to their lives, leading to low motivation.

- **Poor Performance and High Failure Rates:** Outcomes of the G.C.E. O/L exam reflect this disengagement. Each year a large fraction of students fails Mathematics, making it one of the most failed subjects. For example, in one recent examination year, nearly **29% of O/L candidates failed the math exam**. In certain education zones the situation is even worse – studies in zones like Ratnapura indicate that over one-third of students fail math, with overall failure rates in math often exceeding 50% in some cohorts. Such high failure rates have dire implications: students who fail math cannot proceed to many advanced study streams, limiting their future education and career opportunities. This trend suggests that traditional methods are not effectively teaching the material to a significant portion of students.

- **Pedagogical Deficiencies – Rote Learning Over Understanding:** The current system heavily emphasizes preparing for exams through repetitive practice and memorization.

While practice is important, the **exam-oriented culture** has led teachers to prioritize drilling past paper questions and procedural techniques at the expense of developing true conceptual understanding. Topics like algebra, ratios, or area calculation are learned as mechanical procedures. Many students can plug numbers into formulas but do not deeply understand why methods work or how to apply concepts to novel problems. They struggle with problem-solving and higher-order thinking. This gap becomes evident in unfamiliar or applied questions, where students often cannot transfer their knowledge. The lack of innovative pedagogical approaches means students who don't respond to the standard method have little alternative pathways to learn.

- **Insufficient Resources and Technological Gaps:** Another issue is the disparity in resources among schools. Urban schools sometimes have computer labs or multimedia facilities, but many rural schools have limited access to technology. Nationwide, only about 22% of households have computers and reliable internet, which means many students have low exposure to digital learning tools. During the COVID-19 pandemic, these infrastructure gaps resulted in significant learning loss for math, as remote teaching was not effective for all. Even now, most classrooms do not use educational software or interactive content for math due to lack of equipment or training. This digital divide means innovative tools like VR could face adoption challenges if not addressed (a point considered under operational feasibility). However, it also means there is an untapped opportunity to introduce technology where none currently assists learning.
- **Curriculum and Cultural Disconnect:** The national math curriculum, while comprehensive, is delivered in a way that often feels disconnected from students' real-world experiences. Math is taught as a series of abstract topics (fractions, equations, theorem proofs) without linking them to practical examples that Sri Lankan students can relate to. Moreover, most modern educational software or platforms for math (when used) are imported products designed for foreign curricula, often in English only. Language barriers and unfamiliar contexts (e.g. word problems about snow or dollars) can alienate local students. There is a lack of content that situates math in Sri Lankan settings – for instance, using examples from local trade, architecture, or agriculture. As a result, students fail to see the relevance of math in their own heritage and daily life, further dampening interest. Educators have pointed out that weaving cultural narratives into math problems can improve engagement and retention, but this approach is not yet common in the standard pedagogy.

In summary, the problem can be defined as follows: **Sri Lankan O/L math education is hindered by low student engagement, an absence of interactive and contextual learning experiences, and an over-reliance on rote teaching methods**. This has led to unsatisfactory learning outcomes and a disconnection between mathematical concepts and their practical significance. The feasibility study proceeds on the premise that addressing these issues requires an innovative solution that makes mathematics learning active, immersive, and culturally relevant.

Project Description

Lakmaga VR Mathematics Module: The proposed solution to the above problems is a gamified virtual reality module that teaches core Grade 10–11 mathematics concepts through interactive challenges set in historical Sri Lankan contexts. By leveraging VR technology, the module will provide an **experiential learning environment** where students can engage with math concepts in a hands-on manner. The key idea is to situate abstract mathematical problems within **meaningful stories and scenarios** drawn from Sri Lanka's rich heritage – effectively turning math lessons into a series of virtual "quests" or puzzles that students solve by applying mathematical knowledge.

Solution Concept: In this VR experience, students don a standalone VR headset (e.g. Meta Quest) and enter a 3D virtual world that recreates elements of ancient Sri Lankan life, especially those related to agriculture and engineering. Within these environments, they encounter challenges that require using mathematical reasoning to progress. The module comprises three main interactive scenarios (with potential to add more), each focused on specific math topics:

- **1. Sluice Gate Puzzle (Fractions):** Students find themselves by an ancient irrigation reservoir (wewa) equipped with a *Bisokotuwa*, or sluice gate, used historically to control water flow. In this puzzle, the task is to allocate water to different paddy fields in correct proportions. For example, the student might need to open and close virtual sluice valves such that **a certain fraction of the total water** is diverted to each field. They will be given requirements like "Field A needs 1/3 of the water and Field B needs 1/4" and must figure out how to adjust the gates (with visual water flow feedback) to achieve these fractions simultaneously. Through trial and error in VR (pouring water, adjusting flow percentages), learners visually grasp fraction concepts like common denominators and equivalent fractions. The historical context – a king's irrigation manager allocating water – makes the exercise engaging. This scenario ties into math curriculum topics on fractions, ratio/proportion, and basic algebra (for example, solving how much water remains if certain fractions are allocated). It transforms fractions from abstract numbers into volumes of water the student can see and manipulate.

- **2. Granary Area Challenge (Area and Division):** In this scenario, students are placed in charge of an ancient royal **granary** – a large storehouse for grain. They are tasked with organizing the storage or distribution of grain harvest using mathematics. One challenge might involve **calculating areas**: e.g., determining the floor area of the granary or designing storage compartments. The student might be asked to figure out how many standard-sized grain sacks can fit in a granary floor of given dimensions, effectively applying area = length × width calculations. Another challenge can be dividing a harvested grain stock equally among villages (practicing division and understanding of ratios). The VR environment could allow the student to draw boundaries on the granary floor or arrange virtual markers to measure area and visually see their calculations come to life. They might, for instance, rearrange piles of grain or partition the space and immediately see if their division is correct (too little or too much grain per section). This not only reinforces formulas for area (and possibly volume if considering silo shapes) but also gives a sense of practical problem-solving – akin to Tetris with math, fitting shapes to maximize storage. Culturally, it reflects how ancient communities managed resources, connecting mathematics to real economic decisions of the past.

- **3. Stability and Balance Challenge (Logical thinking & Measurement):** This challenge draws on the physics and logic side of mathematics. Students enter a scene perhaps at a construction site of an ancient structure or a marketplace with a **balance scale** used for weighing goods. One puzzle could involve using a traditional balance scale to measure a specific weight of an item using only certain standard weights – a classic logical puzzle requiring **addition, subtraction, and proportional reasoning**. For example, given weights of 1, 2, and 5 units, how to measure exactly 4 units of grain on the scale? The student in VR can place weights on either side of the scale and see it tip or balance, learning the principle of equilibrium (which is essentially an equation: weight on left = weight on right). Another task might involve building a stable structure (like an irrigation canal or small bridge) by correctly measuring and placing support beams – here they must use unit conversions and measure lengths/angles accurately with virtual tools. If measurements are off, the structure in VR might collapse, providing immediate feedback. This trains logical thinking and careful measurement, showing how math is crucial in engineering stability (a real ancient example being how stupas or bridges required precise calculations). It integrates units, arithmetic, and logic in a tangible way.

Across all these scenarios, **gamification elements** are integrated to motivate learners: points are awarded for completing tasks, immediate feedback is provided for incorrect attempts (with hints or visual cues), and progression badges or narrative rewards (e.g., "Master Irrigation Engineer" title) are given as students succeed. The module would be designed to be completed in segments of 10–15 minutes each, making it suitable for use within a class period or as a supplementary activity. Each VR mission aligns with a specific set of curriculum outcomes (fractions, area, etc.), so teachers can deploy the VR activities when those topics are taught, reinforcing the lessons.

Cultural and Educational Alignment: A unique strength of this solution is how it **blends mathematics with Sri Lankan history and culture**. By solving problems set in ancient times, students also indirectly learn about historical technology – e.g., how our ancestors managed water or trade. This interdisciplinary approach can increase student interest (appealing to their sense of identity and curiosity about the past) and address the concern that history education might be getting less emphasis. It demonstrates the practical value of math in a local context, countering the perception that math is just theoretical. The content will be provided in **bilingual form**: instructions and voice narration in Sinhala (with English subtitles or vice versa), to ensure language is not a barrier. This supports students in Sinhala-medium schools and aligns with national education policies encouraging bilingual proficiency.

Furthermore, by being in VR, the solution inherently supports **learning-by-doing**. Students will not be passive listeners; they actively interact and make decisions. Educational research indicates that such active learning approaches can improve understanding, especially in subjects like math and science. Mistakes in the VR world are safe and reversible, encouraging exploration. For instance, a student can try an incorrect fraction setting on the sluice and immediately see the consequence (fields under-watered or flooded), then adjust – a form of constructive feedback loop that is often impossible in a traditional class.

Prototype and Current Status: As part of the broader Lakmaga initiative, some preliminary prototypes of the interactive environments have been developed. A basic version of the Sluice Gate Puzzle has been created in Unity, demonstrating that water flow can be dynamically controlled by user input and fractions can be visualized (e.g., segments of a reservoir) in VR. Similarly, a sample granary environment with simple area calculation tasks has been modeled. These prototypes, although not fully gamified yet, serve as a proof of concept that the technology works on the target hardware (they were tested on a Meta Quest 2 headset) and that students find the scenarios intuitive. Feedback from an informal trial with a few educators was positive – they noted that students were excited to talk about fractions in terms of sharing water, which suggests improved conceptual linkage. Moving forward, these prototypes will be expanded into fully-fledged learning modules with polished graphics, voice-guided instructions, and embedded assessment.

In the subsequent sections, we analyze the feasibility of implementing this VR solution in detail. This includes evaluating whether the necessary technology and skills are available (Technical Feasibility), how the solution would operate in the actual school context (Operational Feasibility), what demand and impact it might have (Market Feasibility), and whether it is financially viable to develop and deploy (Financial Feasibility). Potential risks are identified along the way, with strategies proposed to mitigate them, ensuring that the **Lakmaga Math VR component** can be rolled out successfully to benefit Sri Lankan students.

Feasibility Analysis

The feasibility of the Lakmaga Mathematics VR project is examined from four key angles: technical, operational, market, and financial feasibility. This comprehensive analysis helps determine whether the project can be successfully executed and sustained.

Technical Feasibility

Technology Platform: The project will utilize the **Unity 3D game engine** for development and target deployment on **Meta Quest** standalone VR headsets. Unity is a widely used platform for VR development with robust support, a rich asset store, and a large developer community. The Meta Quest (Quest 2 or Quest 3) provides inside-out tracking and wireless operation, making it suitable for classroom use without requiring PCs or external sensors. This combination (Unity + Quest) is an industry-standard for educational VR applications. Notably, many successful VR learning tools have been built with Unity, so the development team can leverage existing libraries and frameworks (for physics, user interaction, etc.) to accelerate production. Unity's cross-platform capabilities also mean that the content could be adapted to future devices if needed.

Prototype Validation: From a technical standpoint, the core functionality required for the Math VR module has already been **demonstrated in prototype form**. Early-stage prototypes of the math scenarios (as mentioned in the Proposed Solution) were developed in Unity and tested on a Meta Quest 2 headset. These tests confirmed that key interactions – such as real-time water flow adjustments for fractions, area measurement tools, and balance scale physics – work as intended in an immersive 3D environment. For example, the water flow simulation maintained a stable frame rate and responded to user input without noticeable lag, which is crucial to avoid VR motion sickness and ensure a smooth learning experience. The fact that a working prototype exists for core features is a strong indicator of technical viability. It shows that the team has the capability to implement the envisioned mechanics and that the chosen hardware can handle the content.

Development Requirements: In terms of development effort, creating the full VR module will involve several components:

- **3D Content Creation:** The ancient environments (reservoir, granary, marketplace, etc.) need to be modeled in 3D with sufficient realism to be engaging, but also optimized for performance on the Quest. This requires 3D artists to create models and textures. Fortunately, many relevant assets (like generic landscape elements, character models, etc.) can be sourced from Unity's asset store or open libraries, reducing the need to create everything from scratch. Custom models (e.g., a specific type of sluice gate or traditional weighing scale) may need to be created, but these are manageable in scope.
- **Programming and Interaction Design:** Unity's engine will be used to script the interactive logic (using C#). This includes the math logic (e.g., calculating if the fraction of water allocated is correct, or if the measured weight equals target), as well as game

mechanics (scoring, progression). The programming tasks are well within standard practice – Unity provides physics engines (for simulating water and balances) and UI frameworks (for menus, prompts). The development team will implement trigger conditions (like checking answers) and link them to feedback (visual effects or narration cues). None of these tasks are considered cutting-edge research; they are application of existing methods, which is feasible given the skill set of typical Unity developers.

- **Audio and Voice Content:** Since the module will be narrated and bilingual, recording voice-overs in Sinhala and English is a task to plan for. However, Sri Lanka has resources in this area (even possibly within SLIIT or via media students/professionals). The amount of narration is limited to instructional prompts and contextual story dialogue, which can be recorded relatively quickly. Unity easily integrates audio files for playback during scenes.
- **Testing and Iteration:** Testing will be crucial to iron out any technical issues. VR content must maintain **performance** (targeting at least 72 frames per second on Quest for comfort) and be free of major bugs. The team should allocate time for iterative testing, including in-classroom simulations (multiple headsets running concurrently to see if any interference or network needed). Thankfully, Unity's profiling tools and best practices for VR (like optimizing polygon counts, using baked lighting, etc.) are well documented, and the prototype success suggests no obvious performance bottlenecks.

Hardware Considerations: Each school or pilot site will need a set of VR headsets. The Meta Quest devices are self-contained and relatively robust, designed for public use scenarios. They have on-board storage to hold the module (which is expected to be a few gigabytes at most). Battery life (2–3 hours continuous) is sufficient for several class sessions if recharged regularly. From a technical perspective, using **standalone VR** avoids many complications: no tethered cables, no high-end PCs or phones required. It simplifies setup to essentially just ensuring the headsets are charged and the play area (classroom) is clear of obstacles. The headsets use guardian systems to prevent users from moving beyond a boundary, which is helpful for safety. Additionally, since Quest headsets support **casting**, a teacher can mirror a student's view to a phone or monitor to observe progress, which is a useful operational feature.

A potential technical limitation to note is that VR headsets might become **obsolete** over a span of years as new models arrive. However, the software (built in Unity) can be updated to target newer hardware with minimal changes, as long as the new hardware supports standard APIs. The project's code and assets are reusable, meaning the investment in development is not lost if hardware changes – it would mostly involve recompiling for a new device. Moreover, the current trajectory of VR hardware is that devices are becoming more affordable and powerful, so technical feasibility will only improve over time.

Conclusion (Technical): There are no insurmountable technical barriers identified. The required technology – Unity development tools and VR hardware – is readily available and has been proven in similar contexts. The development skills needed (Unity programming, 3D design) are well within the capability of the project team and collaborators, especially given that a functioning prototype exists as a reference. Table 1 below summarizes the key development components and their feasibility status. Overall, the project stands on solid technical ground: it employs established technology and techniques, and preliminary validation has de-risked many core features. Any technical challenges that do arise (such as performance tuning or minor bugs) are considered routine in VR development and can be managed with proper testing and iterative refinement.

Operational Feasibility

Operational feasibility examines whether the VR math module can be successfully integrated into the real-world environment of schools and sustained in day-to-day use. This involves looking at the stakeholders (students, teachers, school administrators), the processes for using the system, and the infrastructure/support required.

Integration into Classroom: The typical use case would involve a teacher incorporating the VR module as a supplement to regular lessons. For example, when covering fractions or geometry in the syllabus, the teacher might allocate a session for students to try the corresponding VR challenge. Practically, not every student may have a personal VR device (especially in early pilot phases), so a rotational model could be used. A class of 40 students could be divided into smaller groups, with perhaps 5–10 VR headsets available. While one group engages in VR for ~15 minutes, other groups can work on related exercises or receive instruction, and then they rotate. This lab rotation model ensures all students get a turn. It has been used in other contexts (like computer labs) and is operationally feasible if properly scheduled. It will be important to ensure that VR sessions fit into class periods – our module is designed with short, modular activities to accommodate this.

Teacher Training and Involvement: Teachers are a critical factor in operational success. If teachers are uncomfortable with the technology or unconvinced of its value, they may not use it effectively. To address this, the project will implement a **teacher training program** (see Appendix B for a detailed outline). Training workshops will be held to:

- Introduce teachers to VR, letting them experience the module as learners first so they understand its objectives and feel its impact.
- Instruct them on the practical handling of devices (charging, fitting headsets, basic troubleshooting).
- Demonstrate how to integrate the VR activities into lesson plans. For instance, a teacher guide would specify what discussion to have before and after the VR session to connect the experience with formal learning objectives (e.g., debrief questions like "How did the water sharing relate to adding fractions?").

- Cover classroom management strategies for VR: ensuring all students are engaged (perhaps having observers or worksheet tasks for those awaiting their turn), and safety precautions (students remain seated or in a clear area while in VR to avoid collisions).

From a feasibility perspective, these training needs are straightforward. Many teachers in Sri Lanka may not have used VR before, but they are generally enthusiastic about tools that can improve student outcomes if given proper support. The content will be provided in Sinhala as needed to ensure understanding. By involving teachers early (possibly even in content testing phases), we can build buy-in. As noted in feasibility studies of similar projects, with adequate training, teachers can become strong advocates and facilitators of VR-based lessons.

Operationally, it is feasible to have one or two teachers at a school act as the "VR champions" who take primary responsibility for the equipment and mentoring colleagues. This might be the ICT teacher or a forward-thinking math teacher. They can help others and coordinate scheduling the VR kit among classes. Given that the module is modular, not every math lesson uses VR – it might be utilized a few times a term – scheduling is manageable.

Infrastructure and Maintenance: The physical requirements for operation include a small open space in the classroom or a lab where students using VR can move their arms freely (a 2m x 2m area per user is recommended). Desks can be pushed aside temporarily. Schools need to have electricity to charge devices (which virtually all O/L schools do have), and secure storage for the equipment. If a computer lab or library is available, devices could be stored and checked out. Since the Quest headsets can work offline once the content is loaded, **internet connectivity is not required during use**. This is a major plus for operational feasibility in rural areas – the entire module can be self-contained on the device. Only periodic updates would require connecting to a Wi-Fi, which could be done by bringing the devices to a connection once in a while (e.g., an admin's phone hotspot or at the education office).

Maintenance of the devices involves keeping them charged and clean, and updating software occasionally. These tasks can be handled by the designated teacher or lab assistant. The project can provide a simple maintenance manual. Based on manufacturer specs, the devices are quite user-friendly; for instance, they auto-update if online, but can also run on current firmware if offline. Operationally, a potential challenge is the **limited number of devices** in initial deployment – this requires coordination. However, starting with a small number of devices in a pilot is actually operationally easier (one class at a time) and if proven successful, scaling up to more devices per school can be planned.

Student Reception and Safety: It is expected that students will be very excited to use VR. This enthusiasm is beneficial, but needs to be channeled appropriately. There must be clear guidelines: e.g., students should remain seated or standing in one spot while using VR (to avoid bumping into others), and others should not interfere with someone wearing a headset. During training sessions, these norms will be communicated to teachers and students. VR usage time per student will be limited (10-15 minutes at a stretch) both to ensure everyone gets a turn and to avoid eye strain or fatigue. Modern VR is generally safe for short durations, with very few incidents of motion sickness especially in static experiences (our module avoids intense motions). We will include an **orientation** for students on how to use the controller, how to exit if uncomfortable, etc. As a backup, any student who feels dizzy can simply take off the headset – the teacher should monitor and have an alternative task for that student. These measures, which are standard in VR educational deployments, mitigate health or safety concerns and make everyday operation smooth.

Administrative and Timetabling Support: For sustained use, school administrators need to support allotting time and possibly adjusting timetables to incorporate VR sessions. Since the module directly ties into the curriculum topics, it can be slotted into existing lesson plans rather than requiring extra periods. For example, during a double-period of math, 20 minutes could be set for VR without losing overall teaching time, as it replaces part of the exercise time with an interactive exercise. Principals and sectional heads would likely support this if evidence (even qualitative feedback) shows improved student interest or understanding. The Education Ministry's STEM drive also favors using technology, so there is alignment with broader educational goals. A small operational concern might be ensuring **all classes get equal opportunity** if devices are limited – this can be handled by scheduling (e.g., Grade 10 classes use it during term 1 for fractions, Grade 11 in term 2 for geometry revision, etc.).

In a pilot scenario with a couple of schools, the operational load is easily manageable: the project team can closely support those schools, gather feedback, and adjust. If scaling to many schools, a train-the-trainer model can be used, and perhaps regional resource centers could hold a few headsets for schools to borrow. These are future considerations; at feasibility stage, we note that **starting small and proving value** is key to operational expansion.

Conclusion (Operational): The operational feasibility is deemed strong. With thoughtful planning, the VR math module can be woven into the fabric of school routines. Teachers, once trained and convinced of the educational merits, can effectively facilitate the VR sessions. The required resources (space, electricity, storage) are minimal and present in virtually all target schools. The bilingual design and curriculum alignment ensure it fits the local context without disruption. Initial deployment at a small scale (one kit per school) is operationally feasible and allows demonstration of success, after which scaling (more devices, more schools) can be handled with support from educational authorities and possibly economies of scale in device provision. In summary, **no significant operational barriers** have been identified; potential challenges like teacher adaptation and device maintenance have clear mitigation strategies, and early adoption signs are positive.

Market Feasibility

Market feasibility considers the demand for the project, its target user base, stakeholder support, and the broader context that could influence adoption. In the case of an educational project like this, the "market" includes schools (teachers, students, administrators), the education ministry, and potentially parents and wider society that values educational outcomes.

Target Audience and Size: The primary users are **Sri Lankan O/L students (Grades 10–11)** studying mathematics. There is a large and renewable cohort every year. Approximately 300,000+ students sit for the G.C.E. O/L exam annually, of which mathematics is a compulsory subject. This indicates a vast potential reach. Even focusing initially on the subset of students in schools equipped or willing to use VR (perhaps urban and semi-urban schools with better facilities), the numbers are significant. Moreover, the government is keen on improving pass rates in subjects like math due to national concern over failures – indicating a receptive climate for innovations that could help. If the pilot demonstrates improved engagement or scores, many schools would likely be interested.

Value Proposition: The Lakmaga Math VR module offers a **unique value proposition**: it provides localized, curriculum-specific VR content in a field where currently **there is none**. To our knowledge, there are very few VR applications tailored to Sri Lankan educational curricula, and virtually none for mathematics in Sinhala language. This lack of localized educational VR content is a gap in the market that our project squarely fills. Schools have started hearing about VR in education from global trends, but they have not had access to content that matches their teaching needs. Introducing Lakmaga gives them a ready-made solution that is aligned to what they teach and easy to deploy (no need to custom create content themselves). This is a strong selling point to educators and policy makers.

Competitive Landscape: On an international level, there are some VR math learning tools and games, but they are not designed for Sri Lankan students. For instance, some apps let students explore shapes or graphs in VR, but they use English and follow foreign curricula. Traditional non-VR educational software (like GeoGebra, or Khan Academy style apps) exist but again are mostly English-based and 2D. The immersive and bilingual nature of Lakmaga sets it apart. In Sri Lanka, initiatives like smart classrooms and digital whiteboards are more common than VR; thus, in terms of direct competition, there isn't a locally made VR solution for us to compete with. This positions Lakmaga as a **pioneer** in the local market, which is advantageous for capturing interest. However, being first also means we must educate stakeholders on what VR is and why it's beneficial, which is part of our outreach.

Stakeholder Support: For market feasibility, gauging the support of key stakeholders is vital:

- **Students** are expected to be enthusiastic; as digital natives, they tend to embrace technology-driven learning, especially something as novel as VR. During initial demos, we anticipate high excitement which translates to word-of-mouth popularity. Their engagement is actually one of the "products" we deliver (engaged students learn better).
- **Teachers** might be cautious initially, but many are aware of the shortcomings in engagement and would welcome a tool that genuinely helps (provided it's not too difficult

to use). In preliminary discussions (an informal survey of a few teachers), the concept was met with curiosity and cautious optimism – teachers liked the idea of visualizing things like geometry in VR and thought students would pay more attention. The concerns they voiced (about managing devices or aligning with syllabus) are being addressed in our design and operational planning.

- **School Administrators and Policy Makers:** The Ministry of Education and provincial education departments have, in recent years, pushed for improving ICT in education. If we approach them with evidence from a pilot (for example, improved math quiz scores or simply testimonials of increased student interest), they are likely to support expansion. Sri Lanka's education authorities have also emphasized **21st-century skills** and innovation; a project like this can be showcased as a model of modernizing education. It's quite feasible that with successful pilot results, the Ministry could endorse the module or include it in some official capacity (even something like recommending schools to use it, or integrating it into teacher training curricula). Such endorsement would greatly expand the "market" because public schools take cues from ministry guidance.
- **Parents and Wider Community:** Parents primarily care about results and their children's future opportunities. If VR can reduce failure rates or improve grades in math (which we expect through better understanding), parents will indirectly support it. Even beyond grades, seeing their children excited to learn math and talk about historical culture is a positive outcome. One operational consideration is if any parents have misconceptions (for example, fearing VR might be harmful or just a game distraction). This can be mitigated by communication – e.g., a demonstration at a school open day to show parents what it is, and emphasizing it's an educational tool, not mere entertainment. Since VR is novel, such transparency will help in acceptance.

Potential Challenges to Market Adoption: One challenge might be the **cost and resource requirements**, as not every school can afford VR hardware immediately. This is where pilot programs and phased approaches come in (see Financial Feasibility and Recommendations). It's feasible to target initial implementation in a range of school types (one high-performing urban school, one average suburban school, one rural school via a donation or grant of devices) to gather data across contexts. If those succeed, it strengthens the case to sponsors or the government to invest in scaling. Another challenge is the **skepticism of traditional educators** who might question whether VR is just a gimmick. To address this, our evaluation plan includes measuring actual learning outcomes (like comparing test performance of those who used VR vs those who didn't for a given lesson). Showing evidence of improved conceptual understanding or at least equal exam performance with higher engagement will be key. In essence, we aim to prove that VR is not only fun but effective.

Scalability and Future Market: Beyond the immediate O/L math use, the platform has potential to expand (as part of Lakmaga) into other grades and subjects. This cross-topic scalability means the investment in VR hardware could be justified across multiple uses – a point we can use to convince schools/ministries. For example, the same set of headsets could host a science module, a history exploration, etc. In market terms, this broadens the appeal: a school wouldn't be buying into just one subject tool, but a whole new platform for learning. The **competitive advantage** of our approach is precisely this multi-subject, culturally enriched design, which few others can offer.

We also note a potential for collaboration with educational content providers or ed-tech companies for further distribution. If proven locally, the concept could be extended to other countries with similar needs (for instance, other South Asian or developing countries where contextual educational VR is lacking). However, our primary "market" remains Sri Lankan O/L curriculum for now.

Market Risks: One risk is if the novelty wears off – after the first few uses, will teachers continue to use it or will it sit idle? To avoid this, we ensure the content is directly tied to syllabus needs (not extra-curricular). It's not an add-on project that can be ignored; it directly helps teach chapters that teachers must cover. Another risk is technological: if VR tech advances quickly, schools might hesitate to invest in current models. Given the pace of school procurement, it's more likely we work with what's available now, and any future advances (cheaper devices) only make it easier. The core content (the math challenges) will remain valuable and can always be ported.

Conclusion (Market): The market feasibility for the Lakmaga Mathematics VR module appears robust. There is a clear **demand for innovative solutions** to improve math education outcomes in Sri Lanka, and our project uniquely addresses that demand. The user base is large and renewable, stakeholder interest is present (with the need for some awareness building), and there are strong alignment factors with national education goals (improving STEM education, integrating technology, preserving cultural context). The lack of competition in localized VR content gives us a first-mover advantage. While challenges in adoption exist (cost, change management), they are surmountable with strategic pilot successes and evidence-based advocacy. If implemented well, the project could not only capture the "market" of forward-thinking schools but also catalyze a broader modernization of math pedagogy in the country.

Financial Feasibility

Financial feasibility assesses whether the project's costs are reasonable in view of the expected benefits and whether funding can be secured. It encompasses development costs, operational costs, and the potential return on investment (ROI) in terms of educational value (and any monetary savings or gains, if applicable).

Development Costs: One advantage of this project is that it is being undertaken in an academic setting (as a student project/thesis component), which significantly reduces labor costs. Much of the development work (coding, designing) is done by the project developer(s) as part of their academic output. Therefore, **direct financial costs for development are relatively low** – primarily limited to equipment and any external services that might be needed:

- *Hardware for Development:* A VR-ready PC (if not already available in the lab) and a couple of Quest headsets for testing. At SLIIT or similar institutions, these might be provided as part of the research infrastructure or can be borrowed. If not, a mid-range gaming PC (~LKR 200,000) and two Quest 2 headsets (~LKR 120,000 each) would be the main capital expense to build and test the system.
- *Software:* Unity is free for educational and personal use. There might be small costs for specific Unity Asset Store purchases (for instance, a water simulation plugin or some 3D models) – but these are usually in the range of $50–100 each, and the project might only need a few, say LKR 50,000 in total for assorted assets. Thus, no expensive software licenses are needed.
- *Content Creation Services:* If voice actors are hired for narration, a modest budget may be allocated (or use volunteer student narrators to save cost). Similarly, if we need a custom artwork or consultation from a subject matter expert (like a historian to ensure cultural accuracy), there might be token honoraria. We estimate these miscellaneous content costs to be minor (perhaps LKR 50,000–100,000 total).

In sum, the **initial development outlay** could be on the order of LKR 500,000–700,000 (~USD 2,000) if equipment is needed; otherwise much less. This is a one-time cost to produce the module.

Deployment and Operational Costs: Deploying the VR module in schools involves acquiring VR headsets and maintaining them. Currently, a Meta Quest 2 (128 GB) in Sri Lanka costs approximately LKR 130,000. Newer models like Quest 3 are about LKR 150,000+. For a pilot, if we provide 5 headsets to a school, that's about LKR 650,000 (USD ~2,000) per school in hardware. Additional costs include charging equipment (just basic power strips, maybe lockable charging cabinet – small amount), and perhaps spare face masks or hygiene covers for shared use of headsets. We also might allocate funds for teacher training workshops (travel for trainers, printing of guides, etc.), though these can often be supported by educational departments.

Maintenance costs are low – mostly replacement of any damaged units or periodic purchasing of new straps/batteries after a couple of years. We could anticipate maybe 5-10% of hardware cost per year in maintenance reserves (e.g., LKR 65,000 per year per set, for replacing one unit or accessories).

Scaling Considerations: If the project moves beyond pilot to wider implementation, economies of scale may kick in. Bulk purchasing of devices could reduce unit costs. Also, newer devices might become cheaper. It's feasible that a few years down the line, standalone VR headsets could be LKR 80,000 each or less, especially if locally assembled or through educational discounts. We note that these costs, while not trivial in a school budget, are on par with other ICT expenditures – for example, a single multimedia projector can cost LKR 100,000 and many schools have those. A computer lab setup runs into millions of rupees. So, equipping a school with a VR kit for ~LKR 0.7 million is not unimaginable if the value is demonstrated.

Funding Sources: Given this is an educational initiative, the typical revenue or profit model of commercial projects doesn't apply directly. Instead, funding would likely come from:

- **Government/Education Ministry:** If the pilot is successful, a proposal could be made for government funding to roll it out. The justification would be improved results and modernized pedagogy. Government funding could cover device procurement and training.
- **Grants and CSR:** Organizations such as UNESCO, UNICEF, or local foundations might be interested in funding innovative educational projects. Corporate Social Responsibility (CSR) programs of banks or tech companies in Sri Lanka also often sponsor digital education efforts. For example, a bank might donate VR headsets to a set of rural schools as part of a math education upliftment program. We consider it feasible to secure grants if we have pilot data – in fact, showing early success will strengthen grant applications.
- **School Investments:** Elite private schools might directly invest in such technology to stay ahead; however, the target is mainly government schools where central funding is needed. Still, a few well-resourced schools could adopt it at their own cost, which provides additional case studies.
- **Scaling within Lakmaga:** If the Lakmaga platform expands to multiple subjects and gets institutional support, costs could be spread. For example, the Ministry might pilot a full Lakmaga package (science + math) in selected zones, doubling the justification for device costs.

From a **cost-benefit perspective**, the benefits are largely qualitative but with significant long-term impact:

- Students who understand math better and pass their O/Ls have improved life prospects (which has economic benefits in the big picture).
- Teachers gain a new tool that can rejuvenate their teaching experience, potentially reducing the need for extra classes or remedial programs (small cost savings).
- If engagement increases, we might see less need for parents to spend on private tuition (a huge private cost in SL for math); while we cannot guarantee that, even a slight reduction in reliance on tuition because school learning became effective would be a societal win.
- Sri Lanka stands to gain an innovative model that could be exported or scaled, which is an intangible benefit in terms of becoming a leader in EdTech in the region.

We can attempt a rough estimation of **ROI in educational terms**: Suppose a pilot of 100 students using VR sees their math pass rate improve by 10% compared to a control group. That means 10 more students pass who otherwise wouldn't, allowing them to continue education. Quantifying that is complex, but each student who passes gains opportunities (A/L, higher education) that can lead to higher earning potential. Even one student avoiding failure has a cascading benefit in a family. These outcomes are the ultimate "returns" we seek.

From a direct financial view, there is no profit mechanism (we are not selling this to students, it's provided to schools), but cost-effectiveness is key. **Labster**, a provider of virtual science labs, has shown that virtual labs can increase pass rates in STEM subjects, which justifies their cost in institutional adoption. By analogy, if our VR module can increase math pass rates or grades, it justifies the expense of hardware when weighed against the cost of repeated exams, remedial teaching, or students dropping out. Each student failing and repeating is a cost to the system; preventing failures has economic value.

Timeline of Costs: Development costs are upfront (Year 0). Pilot deployment costs (Year 1) for a few schools. If successful, scaling costs would come perhaps in Year 2–3. The hardware has a usable life of, say, 3-4 years, so renewal or upgrades might come Year 5 onward. These cycles can be planned in national budgets or grant cycles. The module software itself, once developed, does not incur significant ongoing costs except occasional updates (which could be done by future student teams or small hired teams).

Risk and Sensitivity: A financial risk is that if technology changes (e.g., a new type of device) we might need additional development. But as noted, our content can be ported relatively cheaply since Unity projects are adaptable. Another risk is if adoption is slower than hoped, some devices might sit idle (wasted cost). We mitigate that by carefully selecting pilot sites with committed staff to ensure usage. The initial scale is intentionally modest to evaluate without heavy sunk cost.

Risk Analysis

Implementing the Lakmaga Mathematics VR component involves navigating several risks across technical, operational, pedagogical, and financial dimensions. It is crucial to identify these risks early and plan mitigation strategies to ensure the project's success. Below we outline the key risks and how to address them:

1. **Technical Risks:** There is a possibility of software bugs or performance issues that could disrupt the learning experience. For example, the VR application might crash, freeze, or certain interactive elements might not respond correctly under heavy use. The likelihood of minor bugs is moderate (as with any software), but critical failures can be kept low with thorough testing. Impact: A technical failure during a class session could interrupt learning and reduce confidence in the tool. Mitigation: Perform extensive testing under conditions similar to actual use – testing the module on the Quest devices with multiple back-to-back sessions to catch memory leaks or performance drops. We will have a fallback plan for classes: if the VR module fails unexpectedly, the teacher can switch to a regular exercise or demonstration for that day to avoid wasting class time. Additionally, during initial pilot rollouts, having a developer on standby (or quick remote support) means any critical bugs can be patched rapidly. We plan to schedule maintenance updates each term (applying any

patches or improvements during school holidays). Using Unity's established frameworks and following VR development best practices reduces the risk of technical breakdowns significantly.

2. **Hardware Risks:** VR headsets and controllers are physical devices that can malfunction, get damaged, or become obsolete over time. The risk of device failure or damage (e.g., a student dropping a headset, or battery issues) is fairly high in any classroom setting. Impact: If devices break or underperform, the number of available units could drop, potentially derailing a planned VR session if not enough working headsets remain. Mitigation: We will implement proper training on device handling for both teachers and students to minimize accidental damage. Simple precautions like ensuring the headstrap is secure, using the wrist straps on controllers, and instructing students to remain seated (so they don't bump into walls) will be enforced. We will also provide protective accessories such as silicone covers or foam face padding which not only improve hygiene but add some protection. Keeping a small inventory of spare parts or even an extra headset in each kit is recommended – for example, if we deploy 5 headsets, we might keep a 6th as backup (this is factored into cost contingencies). For the long term, we will plan for hardware refresh cycles: e.g., budgeting to replace or upgrade headsets every 3-4 years as technology advances or units wear out. By monitoring device health and having spares ready, we can prevent a single hardware failure from interrupting class plans.

3. **User Experience & Health Risks:** Using VR can cause motion sickness or eye strain in some users, especially if not designed well. Although our module is mostly stationary and optimized for comfort, a subset of students (perhaps 5-10%) could experience dizziness or nausea. There is also the consideration of general discomfort like headaches if overused. Impact: A student feeling sick can be unpleasant and might create concern among other students or parents about VR's safety. Mitigation: We are designing the VR experiences with comfort in mind – avoiding rapid or unnatural motion in the virtual environment. Teleportation or slow movement will be used instead of fast flying or spinning elements. We will limit session durations to about 10-15 minutes for beginners, which is well within tolerable use. Before the first use, we will screen for any students with known issues (e.g., a history of epilepsy or severe motion sickness) and either have them opt out or proceed with extra caution. Students will always have the option to remove the headset immediately if they feel any discomfort. Classrooms should be well-ventilated and at a comfortable temperature because wearing a headset can get warm. We will include short breaks between VR sessions if a student is doing more than one. Education on using VR will be provided: we'll tell students tips like "if you start to feel dizzy, focus on a fixed point or take the headset off slowly". By following these measures, we expect health-related incidents to be minimal. Additionally, demonstrating to parents and school nurses that VR is being used responsibly will mitigate safety concerns.

4. **Curriculum Alignment & Efficacy Risks:** There is a risk that the VR module might not perfectly align with evolving curriculum or that, despite engagement, it might not directly translate to better exam performance. Syllabi can change over time; if our content isn't updated accordingly, its relevance could diminish. Also, teachers might worry that time spent in VR is time taken from exam preparation, potentially affecting results if the VR learning doesn't cover the right material. Likelihood: Medium over the long term – minor misalignments could grow if curriculums are revised after a few years, and skepticism could rise if exam scores don't show improvement. Mitigation: We plan to maintain close alignment with the curriculum by involving curriculum experts during development and reviewing any syllabus changes annually. The module content can be relatively easily tweaked or expanded via software updates if, say, a new topic is added to O/L math or an existing topic shifts focus. As for demonstrating efficacy, we will integrate some assessments into the VR module (e.g., a quick quiz at the end of each scenario). These scores can be shared with teachers so they see tangible outcomes (for example, a student got all fraction questions right in VR, indicating comprehension). We will also conduct a controlled study in the pilot: one class uses VR and another doesn't for a particular topic, then compare their test results. If results show equal or improved performance for the VR group, it validates the approach. We will document case studies of student learning gains and share these with educators to build confidence that VR is not just "fun" but effective. Essentially, by proving that VR either maintains or boosts test performance, we mitigate the risk that teachers feel it's detracting from exam prep. Continual feedback loops (teacher surveys, student interviews) will also help us refine the content to better meet learning objectives.

5. **Teacher Adoption and Training Risks:** If teachers do not buy-in to the VR module or feel unprepared to use it, the project could falter. This risk is moderate – some teachers may be hesitant about new technology or worried it's too complex to manage. Since teachers are gatekeepers in the classroom, their reluctance could mean the VR kit stays in a cupboard unused. Mitigation: As discussed in Operational Feasibility, providing comprehensive training and ongoing support is key. Training sessions will be hands-on, allowing teachers to experience the student perspective in VR and guiding them step-by-step on operating the system. We will supply easy-to-follow guides (with screenshots, troublehooting tips, and lesson integration ideas). Another strategy is to identify or appoint a "tech champion" teacher at each school – someone enthusiastic about the project who can assist others and lead by example. Often seeing a peer successfully use the tool and achieve good results with students will encourage other teachers to try it. Starting the implementation with willing and enthusiastic teachers (rather than forcing all to start at once) will generate early success stories and positive word-of-mouth in the staffroom. We will also actively involve teachers in the pilot evaluation – their feedback will be sought and visibly incorporated into improvements, so they feel ownership. By building a community of practice (possibly through a chat group or forum for teachers using Lakmaga

to share experiences), we reduce the psychological barriers and create a support network. As teachers see students benefiting and realize that using VR is not as daunting as it first appears, their adoption will increase.

6. **Financial and Sustainability Risks:** Funding availability is a risk; if after the pilot there is no budget to continue or expand, the project could stall. Also, devices require periodic replacement – a lack of funds in the future could hamper the program's longevity. Mitigation: We have planned for engaging stakeholders (Ministry, NGOs, private sector) early to secure buy-in for funding (as detailed in Market and Financial feasibility). Demonstrating strong pilot results is the best mitigation – success will attract funding. We will prepare a sustainability plan for schools, including possibly low-cost maintenance models (like a small annual fee for software support if needed, or encouraging schools to allocate part of their ICT budget for VR upkeep). Another mitigation is exploring shared resource models: perhaps a set of VR kits managed by the Zonal education office that circulates among schools – this requires coordination but can sustain usage if individual school budgets are tight. On the flip side, as hardware costs are expected to decline, adopting VR may become cheaper over time, reducing financial strain. By showing cost-effectiveness (e.g., improved pass rates which means fewer repeaters – saving government expenditure on repeat exams/years), we can make a case that the program pays for itself in broader terms.

In conclusion, the risk analysis reveals that while there are multiple risks to manage, none of them are insurmountable. Each identified risk comes with a clear mitigation strategy, and many of these have been informed by experiences from similar educational technology projects (including the Science component of Lakmaga). By proactively addressing these risks – through rigorous testing, training, stakeholder engagement, and contingency planning – we can greatly reduce their likelihood and impact. This means the project can proceed with a strong chance of success. Far from being deterrents, these risk considerations have helped us strengthen the project plan: we have built a more robust, teacher-friendly, and well-supported initiative as a result of this analysis. Ongoing monitoring of these risk areas will be part of project management, ensuring that any new issues are caught early and handled. Overall, with the mitigation measures in place, the potential benefits of the Math VR module far outweigh the manageable risks involved in its implementation.

Recommendations

After conducting the above feasibility analyses, this study finds that the Mathematics component of the Lakmaga VR learning platform is viable and holds significant promise for improving math education in Sri Lanka. To ensure the successful implementation and maximization of the project's impact, the following recommendations are put forward:

1. **Proceed with a Phased Pilot Implementation:** It is recommended to move forward with development completion and deploy the VR module in a controlled pilot program. Start with a small number of volunteer schools (for example, 2–3 schools representing different contexts such as urban and rural) before scaling up. This phased approach will allow the project team to observe real classroom usage, gather feedback, and measure outcomes in a manageable setting. The pilot should run for at least one term, covering several math topics, so that sufficient data on student performance and engagement can be collected. Based on pilot results, refine the content and address any issues, then expand to more schools in Phase 2. This cautious scaling ensures any unforeseen challenges are ironed out early and demonstrates proof-of-concept to potential funders for broader rollout.

2. **Secure Stakeholder Endorsement and Collaboration:** Engage with the Ministry of Education and provincial education authorities from the outset of the pilot. Having official endorsement can facilitate adoption and resource allocation (like teacher training days or funds for devices). We recommend forming an advisory panel or working group including a representative from the National Institute of Education (NIE), a few experienced math teachers, and a curriculum specialist. Their input will ensure the VR content remains aligned to educational standards and can pave the way for integrating the module into mainstream practices if successful. Additionally, collaborating with organizations such as the National e-Learning Resource Center or university education faculties could provide valuable pedagogical insights and help in teacher training efforts.

3. **Conduct Ongoing Teacher Training and Support:** As highlighted in feasibility sections, teacher readiness is crucial. It is recommended to develop a comprehensive teacher training program specific to the VR module. This includes initial workshops before deployment and periodic follow-ups (e.g., a mid-pilot review session). Training should not be one-off; consider establishing a helpdesk or peer-support network (even a simple WhatsApp group for teachers using Lakmaga VR) where they can quickly get answers or share experiences. By empowering teachers with confidence and competence in using technology, we ensure sustained use. It's also recommended to identify "master trainers" – teachers from the pilot who become proficient and can train colleagues in new schools during scale-up, creating a multiplier effect for training efforts.

4. **Ensure Alignment with Curriculum and Exams:** Continue to closely map all VR activities to the official math syllabus and exam paper formats. We recommend creating a

simple reference guide for teachers that explicitly states which syllabus unit and learning outcomes each VR scenario addresses. For example, “Sluice Gate Puzzle – aligns with Grade 10 Mathematics Unit 5: Fractions (Learning Outcome X)”. Moreover, incorporate exam-style questions into the pre/post assessments of the VR module, so teachers see that it directly prepares students for exams (albeit in a more engaging way). If possible, coordinate with curriculum developers – for instance, share results with them and suggest including references to such VR activities in future teacher guides. This will solidify the module’s role as a credible tool rather than an add-on.

5. **Monitor and Evaluate Learning Outcomes Rigorously:** It is crucial to measure the impact of the VR module on student learning to justify and guide further investment. Implement an evaluation plan (see Appendix C) that uses both quantitative and qualitative metrics. For each topic covered with VR, assess student performance through quizzes or assignments and compare with previous cohorts or control groups. Collect student feedback on their confidence and interest in math after using VR. Also, solicit teacher observations on changes in class participation or understanding. This data should be systematically analyzed. If certain areas show no improvement, tweak the approach accordingly. Document any improvement in exam pass rates or grades in pilot schools. By the end of the pilot, prepare an evaluation report showcasing results – this will be instrumental in convincing stakeholders (government, donors) to support scaling up. Positive evidence will also help get skeptical teachers on board in new schools.

6. **Develop a Sustainability and Scale-Up Plan:** Before moving to large-scale implementation, plan for long-term sustainability. This includes budgeting for hardware maintenance/replacement every few years and updates to software as needed (for example, when the curriculum changes or as technology evolves). Explore funding options: the government might include this in its annual education budget if proven successful, or partnerships with private sector could be formed (e.g., a telecom company sponsoring devices for rural schools as CSR). We recommend preparing a costed scale-up proposal that outlines how to go from 3 pilot schools to 50 or 100 schools over a 3-year period, including training, support, and production of additional content. Part of sustainability is also integrating the module into the regular school system – lobby for inclusion of VR session guidelines in the national mathematics teacher guide, so it becomes part of recommended practice. By planning these aspects early, the project can transition smoothly from pilot to expansion without losing momentum.

7. **Enhance Content Based on Feedback and Extend Cultural Integration:** Use the pilot phase to gather rich feedback on the content. If, for instance, teachers suggest an additional scenario or students find a particular element confusing, incorporate those improvements. It’s recommended to enhance the storytelling and cultural elements further if it increases engagement – perhaps add more historical anecdotes or tie in a bit of folklore to math problems, as long as it doesn’t distract from learning. The uniqueness of Lakmaga is the

cultural angle, so double down on that strength: e.g., include a short segment where after completing a challenge, students learn a fun fact about ancient technology they just used. This cross-curricular bonus (history and math) can attract broader support (maybe even from those concerned about diminishing history education). As new historical sites or concepts come to light, consider if they can inspire future VR challenges (ensuring the module stays fresh and locally relevant).

8. **Promote Inclusive and Equitable Access:** When expanding, ensure that not only elite urban schools benefit but also under-resourced schools. We recommend working with educational authorities to identify a few rural or disadvantaged schools for early inclusion, potentially through sponsored hardware if needed. Demonstrating success in a rural school will strengthen the case that this solution is scalable across varying conditions, and it will prevent a digital divide where only some students get these opportunities. Also, consider language inclusion: currently Sinhala/English is covered – eventually, adding Tamil language support would be important for national equity (perhaps in a second phase). We advise planning for a Tamil translation of the module once the Sinhala/English version is fully tested, so that Tamil-medium schools can also partake. This could involve collaborating with a Tamil-medium math teacher to adapt the narration and text.

9. **Foster a Community of Practice:** Encourage the creation of a community or network of teachers and schools using the VR module (a Lakmaga Math VR user group). This could start informally online (as mentioned with messaging groups) and later evolve into workshops or an annual meet-up to share best practices. For example, teachers could share how they integrated a VR lesson with a follow-up activity or how they managed students who were quick vs. slow to solve the VR puzzles. A community will also help in troubleshooting – teachers can help each other solve minor issues, reducing dependency on the core team. The project team should facilitate this initially by providing the platform (maybe an online forum or periodic newsletter highlighting success stories). In the long run, this community can advocate for the program's continuation and growth, creating grassroots support that complements top-down approval.

10. **Publicize Success and Generate Support:** Finally, it's recommended to document and publicize the pilot's successes to build wider support. This might include creating a short video that shows students using the VR module and talking about what they learned, which can be shared with education sector leaders, at conferences, or on social media. Engage local media to possibly do a feature on the project (e.g., a newspaper article on how a rural school improved math results with VR – human interest + education innovation story). Public support can pressure decision-makers to adopt the program more widely. It may also attract international attention and partnerships (for instance, EdTech organizations or researchers might want to support or study the project). The goal is to make Lakmaga VR Math not just a pilot at a few schools, but a known innovation in Sri Lankan education with momentum behind it.

By following the above recommendations, the project team can increase the likelihood of a successful outcome – one where students gain a deeper understanding of mathematics, teachers gain a valuable new teaching aid, and the education system benefits from an innovative model that merges technology with cultural learning. These steps will help ensure that the Lakmaga Mathematics VR component not only launches effectively but also grows sustainably and contributes meaningfully to the improvement of secondary education in Sri Lanka.

Conclusion

This This feasibility study examined the proposed **Mathematics component** of the "Lakmaga" Gamified VR Learning Platform in detail, evaluating its introduction into Sri Lanka's O/L mathematics curriculum from multiple angles within an ancient Sri Lankan cultural context. The analysis reveals that the project is not only feasible but also timely and potentially transformative for mathematics education.

In the **Introduction** and **Problem Identification**, we established the need for such an intervention: traditional methods have left many students disengaged and underperforming in math, a subject critical for their future opportunities. The lack of interactive and contextually rich learning experiences contributes to high failure rates and a perception of math as an abstract, daunting subject. The **Proposed Solution** addresses these issues by leveraging immersive VR technology to create learning experiences that are interactive, engaging, and rooted in local historical scenarios. By doing so, it makes mathematics tangible and relevant, helping students bridge the gap between theory and real-world application.

Through the **Feasibility Analysis**, we found that:

- **Technically**, the project stands on solid ground. Unity and Meta Quest VR are appropriate and proven tools for this endeavor, and early prototypes validate that the core features can be implemented. No insurmountable technical hurdles were identified – performance can be managed and development uses standard practices. The technology needed is readily available and within the capacity of the development team.
- **Operationally**, integrating the VR module into schools is practicable with careful planning. Teachers can be trained, schedules can be adapted to include VR sessions, and infrastructure needs (power, space, device management) are modest and feasible for most schools. The bilingual content and curriculum alignment ensure it fits the local classroom context. Potential operational challenges, such as device maintenance and teacher adoption, can be effectively mitigated by the strategies outlined.
- **Market-wise**, there is a strong demand and supportive climate for such an innovation. The large number of students and the national focus on improving math outcomes create a ready "market" in schools. The uniqueness of a culturally tailored VR tool gives it a competitive edge, essentially meeting an unfulfilled need. Stakeholders from students up to policy

makers have incentives to support it if it demonstrates results. With proper stakeholder engagement and evidence from pilots, scaling up is very feasible.

- **Financially**, the project requires a moderate initial investment, primarily in hardware, but promises high returns in educational value. The costs are justified by the potential benefits, such as improved pass rates and enhanced student skills, which have long-term socio-economic payoffs. Moreover, funding avenues (government, grants, CSR) are available to cover these costs, especially once the concept is proven effective. The analysis shows that maintaining and expanding the project is financially manageable within the scope of typical educational expenditures.

The **Risk Analysis** did not uncover any “showstopper” risks; rather, it enumerated challenges that can be managed through proactive measures. From technical glitches to teacher adoption, each risk has a mitigation plan in place. For instance, thorough testing and backup plans address technical issues, training and support address human factors, and alignment and assessment plans address curriculum concerns. By being mindful of these risks and addressing them early, the project can avoid common pitfalls and ensure smooth implementation.

In light of these findings, we presented a set of **Recommendations** to guide the next steps. Key among them is to undertake a phased pilot deployment, closely monitor outcomes, and engage stakeholders at all levels to build support. Emphasizing teacher training and community-building will be crucial for sustainability. Additionally, documenting success and maintaining curriculum relevance will ensure the project remains effective and valued. These recommended actions form a roadmap for moving forward from feasibility study to actual practice.

In conclusion, the Mathematics component of the Lakmaga VR platform is a **feasible and highly promising project**. It aligns well with educational needs and goals in Sri Lanka, employs suitable and available technology, and can be realized with the resources and strategies identified in this report. We conclude that the project should move into the implementation phase, using the pilot approach and recommendations as a guide. With diligent follow-through on training, evaluation, and stakeholder engagement, this initiative can significantly enhance math education – making learning more interactive, enjoyable, and effective for Sri Lankan students.

By transforming mathematics from a dry, abstract subject into a living experience of problem-solving rooted in the nation’s own heritage, Lakmaga has the potential to rekindle students’ interest and confidence in math. If successful, it will not only improve performance in the short term but also instill a deeper understanding and appreciation of mathematics as both a practical tool and a part of cultural legacy. The feasibility study supports taking this important step towards modernizing education: bringing together the old and the new – ancient wisdom and cutting-edge technology to light the path for future generations of learners.

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